



Adverse effects on sexual development in rat offspring after low dose exposure to a mixture of endocrine disrupting pesticides

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ABSTRACT

The present study investigated whether a mixture of low doses of five environmentally relevant endocrine disrupting pesticides, epoxiconazole, mancozeb, prochloraz, tebuconazole and procymidone, would cause adverse developmental toxicity effects in rats. In rat dams, a significant increase in gestation length was seen, while in male offspring increased nipple retention and increased incidence and severity of genital malformations were observed. Severe mixture effects on gestation length, nipple retention and genital malformations were seen at dose levels where the individual pesticides caused no or smaller effects when given alone. Generally, the mixture effect predictions based on dose-additivity were in good agreement with the observed effects. The results indicate that there is a need for modification of risk assessment procedures for pesticides, in order to take account of the mixture effects and cumulative intake, because of the potentially serious impact of mixed exposure on development and reproduction in humans.

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1. Introduction

The high prevalence of disorders related to the endocrine system, e.g. hormone-dependent cancers, fertility problems and congenital malformations of reproductive organs, is a point of increasing concern in the western world [1,2]. Many genetic, environmental and lifestyle factors may be involved in these adverse effects, one of them possibly being exposures to endocrine disrupting chemicals (EDCs).

Currently, risk assessment of chemicals including pesticides is based on the no observed adverse effect levels (NOAELs) for effects of single compounds. Based on results from animal studies, exposure to single endocrine disrupting chemicals (EDCs) generally does not cause concern for adverse reproductive effects in humans. Humans are, however, exposed to a mixture of several EDCs [3,4], and during the last decade, scientific and regulatory focus has gradually begun shifting towards examining the effects of mixtures.

Since 2005 the European Union member states have for example been obliged to evaluate and if possible refine existing methodologies in order to take combined actions of pesticides into account during risk assessment and especially when establishing maximum residue levels (MRLs) [5].

In studies where experimental animals have been exposed to several endocrine disrupters simultaneously, substantial mixture effects on reproductive development have been seen even though each of the individual EDCs were present at low, ineffective doses [6–8]. In addition, there are indications that cumulative exposure to EDCs including pesticides may play a role in relation to effects on human development, as epidemiological studies have reported increased prevalence of cryptorchidism in sons of women working as gardeners [9] or living on farms where pesticides have been used [10,11]. Furthermore, epidemiological studies have found associations between cryptorchidism and hypospadias and total estrogenic load measured in mother's placenta extracts [12], and association between congenital cryptorchidism and levels of certain organochlorine pesticides in breast milk [13]. Swan et al. [4] found that decreases in anogenital distance among male infants are associated with prenatal phthalate exposure, and Pierik et al. [14]

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identified paternal exposures to pesticides and smoking as factors associated with these congenital malformations. Thus, it is important to bear in mind that exposures to endocrine disruptors may contribute to a combined adverse effect, even though the effects of the single compounds are undetectable [6,8,15]. The findings from animal mixture studies should have major implications for the human risk assessment of EDCs, as they imply that the current use of NOAELs for single chemicals may lead to an underestimation of the potential risk for humans exposed to mixtures of chemicals.

A wide range of pesticides is suspected to act as endocrine disruptors, and there can be many different mechanisms causing the observed effects in animal studies. The five presently investigated pesticides, epoxiconazole, mancozeb, prochloraz, tebuconazole and procymidone, were selected based on information on pesticide use in Denmark and within the EU, and on their endocrine disrupting activity and effects. Procymidone competitively antagonizes the binding of androgens to the androgen receptor, and thereby mainly affects the reproductive development in male offspring [16,17]. Common features of the azole fungicides epoxiconazole, prochloraz and tebuconazole are that they increase gestational length in dams, virilize female pups and affect steroid hormone levels in fetuses and/or dams [18–20]. Mancozeb acts mainly *via* disruption of the thyroid hormone system and is therefore suspected of disrupting brain development [21,22].

The applied extended developmental toxicity model in rats has previously been employed by our group to characterize the effects of developmental exposure to EDCs on male offspring, and detailed mixture experiments have revealed that statistically significant mixture effects can be observed when anti-androgenic chemicals are combined at levels below their individual NOAELs [6,7]. Before initiating the experimental study presented here, the mixture ratio and dose levels of the individual pesticides were chosen based on earlier obtained dose–response relationships for each of the five pesticides. Hereafter, range-finding studies were performed to find a combined dose that represented the pesticides at doses close to their individual NOAELs, without causing major effects on the pregnant animals or on pup survival. Data from these range-finding studies are presented in Jacobsen et al. [23] and were used to set the dose levels of the mixture and of the individual compounds. Three different doses of the pesticide mixture and two doses of the individual pesticides were investigated in pregnant rat dams, and gestation length as well as anogenital distance, nipple retention and incidence of congenital malformations in the offspring were registered. Based on single chemical dose–response curves obtained in both previous and the present study, the effects of the pesticide mixture were predicted using the mathematical models dose-addition and independent action. Other endpoints sensitive to endocrine disruption, such as reproductive organ weights and histopathology, semen quality and a wide range of neurobehavioral endpoints were also measured in the study, and are presented in Jacobsen et al. [24]. Furthermore, chemical analysis of the pesticides in blood samples from dams and offspring were used for evaluating if mixture exposure and exposure to the single pesticides would lead to comparable blood levels. These results are presented in Herrmann et al. [25].

One goal of mixture toxicology is to estimate the toxicities of untested mixtures of chemicals on basis of information on individual components. For predicting combined effects of the five pesticides, we assumed that each pesticide in the mixture did not exacerbate or diminish the effects of the other pesticides. The choice of an appropriate model for calculation of additivity expectations is essential for assessments of mixture effects, because it is in relation to these additivity expectations that combination effects are judged in terms of synergisms or antagonisms [26]. Mixture effects according to “additivity” assumptions can be calculated

by using two alternative concepts, dose addition and independent action. Dose addition looks at mixture effects in terms of a “dilution principle”. It assumes that one chemical can be replaced totally or in part by an equal fraction of an equi-effective dose of another, without diminishing the overall combined effect [27]. Dose addition is often used for mixtures composed of chemicals that act through a similar or common mode of action [28–32]. Its application to the present mixture of five pesticides appeared justified because all pesticides are endocrine disruptors and – apart from mancozeb – produce a common effect outcome, *i.e.* anti-androgenic effects in male offspring. However, it can be argued with equal justification that the similarity assumption for dose addition is not applicable to the chosen mixture because the pesticides produce their effects by a diversity of molecular mechanisms. For this reason, we also employed the alternative concept of independent action to estimate additivity expectations. Independent action assumes that the joint effects of a combination of agents can be calculated by adopting the statistical concept of independent events [33]. It is viewed as appropriate for mixtures of chemicals with diverse modes of action [28], however, dose addition and independent action often yield additive mixture effect predictions within the same range.

The overall aim was to explore whether a mixture of environmentally relevant endocrine disrupting pesticides with dissimilar modes of action would cause adverse developmental toxicity effects in rats at dose levels below NOAELs for the individual pesticides. Furthermore, the aim was to investigate whether dose-additivity or independent action predictions of the expected mixture effects resulted in useful estimates compared to the observed mixture effects for relevant endpoints. The mixture and the single pesticides were also investigated using *in vitro* assays, in order to compare these results with those from the rat study, and evaluate the usability of alternative *in vitro* methods for estimating potential mixture effects.

2. Materials and methods

2.1. Chemicals

The five pesticides used were epoxiconazole (CAS no. 106325-08-8, purity 99.0), mancozeb (CAS no. 8018-01-7, purity 76.0), prochloraz (CAS no. 67747-09-5, purity 98.5), tebuconazole (CAS no. 107534-96-3, purity 98.5) and procymidone (CAS no. 32809-16-8, purity 99.5). All chemicals were purchased in a technical quality from VWR-Bie & Berntsen, Herlev, Denmark. Corn oil (Sigma–Aldrich, Brøndby, Denmark) was used as vehicle.

2.2. Animals and exposure

The animal studies were performed under conditions approved by the Danish Animal Experiments Inspectorate and by the in-house Animal Welfare Committee.

The study included 198 time-mated nulliparous, young adult animals (Han-Tac:WH, Taconic Europe, Ejby, Denmark) distributed to 14 groups of animals. It was performed in 4 blocks with a week between each block and the 14 groups were as equally as possible distributed among the 4 blocks. The animals were housed in pairs until GD (Gestation Day) 18 and alone thereafter under standard conditions in semi-transparent polycarbonate cages (15 cm × 27 cm × 43 cm) with Aspen bedding (Tapvei, Denmark) situated in an animal room with controlled environmental conditions (12 h light–dark cycles with light starting at 9 p.m., light intensity 500 lx, temperature 21 ± 2 °C, humidity 50% ± 5%, ventilation 8 air changes per h). A complete rodent diet for growing animals ALTROMIN 1314 (Soy- and alfalfa-free ALTROMIN GmbH, Lage, Germany) and acidified tap water (to prevent microbial growth) was provided *ad libitum*. The animals were observed twice daily for signs of toxicity and body weights were recorded daily during the dosing period.

On the day after arrival, at gestation day (GD) 4, the time-mated animals were pseudorandomly distributed into groups with similar body weight (bw) distributions. They were given four days after arrival to adapt to the reversed light–dark cycle before beginning the exposure. Independently of actual day of delivery, the expected day of delivery, GD 23, was designated pup day (PD) 1 for the pups. Thereby, the age of the pups related to the time of conception, but was rather similar to postnatal age. Dams were dosed daily by gavage, from GD 7 to 21 and from the day after birth to pup (PD) 16, and were treated at a constant volume of 2 ml/kg/day, with individual doses based on the body weight of the animal on the day of dosing.

Table 1

Doses of the pesticides administered individually and in mixture to pregnant rat dam from GD (gestation day) 7 to PD (pup day) 16 (in mg/kg bw/day) and number of dosed time-mated rats (dams with viable litters) in each group.

Group	n ^a	Epoxiconazole	Mancozeb	Prochloraz	Tebuconazole	Procymidone	Pesticide mixture
1: Control	22 (15)	–	–	–	–	–	–
2: Pestimix-14.6	22 (17)	1.25	2.08	2.92	4.17	4.17	14.6
3: Pestimix-29.2	22 (9)	2.5	4.17	5.83	8.33	8.33	29.2
4: Pestimix-43.8	22 (14)	3.75	6.25	8.75	12.5	12.5	43.8
5: Epoxi-3.75	12 (8)	3.75	–	–	–	–	–
6: Epoxi-15	10 (4)	15	–	–	–	–	–
7: Manz-6.25	12 (5)	–	6.25	–	–	–	–
8: Manz-25	10 (7)	–	25	–	–	–	–
9: Prchl-8.75	12 (8)	–	–	8.75	–	–	–
10: Prchl-35	10 (4)	–	–	35	–	–	–
11: Tebu-12.5	12 (8)	–	–	–	12.5	–	–
12: Tebu-50	10 (6)	–	–	–	50	–	–
13: Procy-12.5	12 (7)	–	–	–	–	12.5	–
14: Procy-50	10 (4)	–	–	–	–	50	–

Epoxi: epoxiconazole; Manz: mancozeb; Prchl: prochloraz; Tebu: tebuconazole; Procy: procymidon.

^a The large difference between the number of exposed time-mated rats and dams with viable litters was mainly due to unusually low pregnancy rate from the animal breeders and not to high rates of pre- and postnatal mortality in the offspring.

2.3. Mixture composition

To design the mixture of the five pesticides, developmental toxicity data from previous studies with each of the pesticides were used. The mixture ratio for the five pesticides was chosen based on the dose–response relationship and the NOAEL for effects on increased gestation length in dams and perinatal mortality in the offspring in previous studies in our laboratory (Supplemental Material Table X1). Epoxiconazole, prochloraz and tebuconazole cause effects on pregnancy length and perinatal survival. The effects on perinatal mortality at the LOAELs were more marked for epoxiconazole and tebuconazole than for prochloraz. Based on this the doses chosen for epoxiconazole and tebuconazole in the mixture were their NOAELs for these effects, whereas a slightly higher dose of 35 mg/kg/day was chosen for prochloraz. Neither mancozeb nor procymidone caused effects on pregnancy length or perinatal survival at the doses studied. For mancozeb, a dose level of 50 mg/kg/day in the mixture was initially chosen, because our earlier studies have shown that dose levels of 150 mg/kg/day and higher cause marked acute neurotoxicity in pregnant animals [21]. Due to acute neurotoxicity in a range-finding study using non-pregnant animals the mancozeb dose was decreased to 25 mg/kg/day in all of the studies in pregnant animals. The dose level for procymidone was chosen mainly based on the anti-androgenic potency in our earlier studies (Supplemental Material Table X1).

Upon choosing the mixture ratio, two range-finding studies were performed. The first was in non-pregnant animals while the second was in pregnant animals, and they were performed in order to test for toxicity and endocrine disrupting effects of various mixture doses [23]. The highest dose of the mixture, which did not cause maternal toxicity in the range-finding study in pregnant animals, was used as starting point for the present study.

In Table 1 the composition of the pesticide mixture, the doses of the pesticides administered individually and in mixture and the number of litters in each group are shown. Four groups of 22 time-mated rats were given daily oral doses of 0, 14.6, 29.2 or 43.8 mg/kg/day of the mixture of the 5 pesticides, whereas ten groups of 10 or 12 time-mated rats were similarly dosed with two doses of the individual pesticides. The lowest dose of each pesticide was similar to the dose included in the highest mixture dose and the highest dose of the single pesticides was 4 times higher.

2.4. Gestation length, anogenital distance and nipple retention

After delivery, which occurred from GD22 to 24, the weights of dams and individual pups were recorded and the pups were counted, sexed, checked for anomalies and anogenital distance (AGD) was measured using a stereomicroscope. Additionally, anogenital index (AGDI), i.e. AGD/cubic root of body weight was calculated for all offspring. Pups found dead were macroscopically investigated for changes when possible. Weight of pups was recorded again on days 6, 13, 22. On PD 13, all male and female offspring were examined for the presence of areolas/nipples (nipple retention, NR), described as a dark focal area (with or without a nipple bud) located where nipples are normally present in female offspring.

2.5. Genital malformations in male offspring

On PD 16 and PD 22, 1–3 male pups per litter were randomly selected for autopsy. On PD 22 the pups were decapitated in CO₂/O₂ anesthesia. Male reproductive organs were examined macroscopically for anomalies and testicular descent and scored for the degree of external genital malformations on a scale from 0 to 3, with the observer being blinded with respect to dose group.

The scores were:

Score 0 (no effect): Normal genital tubercle, with the urethral opening found at the tip of the genital tubercle and the preputial skin intact.

Score 1 (mild dysgenesis of the external genitals): A small cavity on the inferior side of the genital tubercle or a minor cleft in the preputial opening was observed, estimated 0.5–1.4 on an arbitrary scale. The size of the genital tubercle was decreased.

Score 2 (moderate dysgenesis of the external genitals): The preputial cleft was larger, estimated 1.5–2.4 on an arbitrary scale. The urethral opening was situated half-way down towards the base of the genital tubercle (hypospadias).

Score 3 (severe dysgenesis of the external genitals): The preputial cleft was large, estimated 2.5–3.5 on an arbitrary scale. The urethral opening was situated further than half-way down the inferior side of the genital tubercle to the base of the genital tubercle (hypospadias). At the base of the genital tubercle a groove extending laterally was observed (similar to control females).

Gubernacular length was determined as a measure of testicular descent, as an increased gubernacular length would indicate cryptorchidism.

All weaned male offspring were scored alive for external genital malformations shortly after sexual maturation (around PD 50) using a similar scoring system, that was modified to reflect the development of the external male sex organs.

At the age of approximately 9 months, the 1–2 males per litter were decapitated in CO₂/O₂ anesthesia, and external genitals were inspected for the following anomalies: descendent testes, alopecia in the perineal area, cleft phallus and hypospadias. Malformations were scored using a system with 0 denoting the normal and 3 the most severe changes. The observer was blinded with respect to dose groups.

2.6. Statistics

For all analyses, the alpha level was set at 0.05 and the litter was the statistical unit. Data were examined for normal distribution and homogeneity of variance, and if relevant, transformed. In cases where normal distribution and homogeneity of variance could not be obtained by data transformation, a non-parametric Kruskal–Wallis test was used, followed by Wilcoxon's test for pair wise comparisons. Data with normal distribution and homogeneity of variance were analyzed using analysis of variance (ANOVA). When more than one pup from each litter was examined, statistical analyses were adjusted using litter as an independent, random and nested factor in ANOVA or litter means were used. Where an overall significant treatment effect was observed, two-tailed comparison was performed using least square means or Dunnett. Birth weights were analyzed using the number of offspring per litter as covariate and AGD was analyzed using body weight as a covariate. Additionally, anogenital-index (AGDI), i.e. AGD divided by the cube root of body weight, was analyzed. The cube root was used because this converts a three-dimensional end point (weight) into a one-dimensional such as the AGD [34].

When analyzing the level of dysgenesis or malformations of external male organs the scores were categorized into a binary variable with score 0 (normal) and scores 1, 2 and 3 (mild to severe effect). The results are generally shown both as number of offspring and number of litters affected per group, because malformations are rare events that may only affect very few litters. Statistical analysis based on the litter as unit is therefore rather insensitive and consequently it is current practice also to include offspring data in the toxicological evaluation. The statistical analyses of the malformations were done using Fisher's exact test on litter data.

The number of nipple/areolas was assumed to follow a binomial distribution with a response range between 0 and $\theta_{\max}$, with $\theta_{\max}$ being equal to the biologically possible maximal number of nipples in rats, either 12 or 13. The choice of $\theta_{\max}$ was decided on considering the global fit (information criterion of Schwarz). To account for litter effects on NR, correlation structures between number of nipple/areolas and litter were modeled by the Generalized Estimating Equations method as in Hass et al. [6]. The number of nipples in the all control group males was zero and it was therefore necessary to put in 1 nipple in 3 pups from the control group to perform

the statistical model. All statistical analysis was performed using the SAS procedure PROC GENMOD.

Asterisks in tables and figures, indicate a statistically significant difference compared to controls * $p \leq 0.05$; ** $p < 0.01$; *** $p < 0.001$. All analyses were performed using SAS Enterprise Guide 3.0, SAS Institute Inc, Cary, NC, USA.

2.7. Mixture modeling

The data for the single pesticides comprised the raw data from our previous studies (overview in Supplemental Material Table X1) and the data for both single pesticides and the mixture obtained in the present study.

Endpoints considered relevant for this modeling were those observed to show an effect at mixture dose levels at which the individual pesticides when given alone caused no or significantly smaller effects. These endpoints included gestation length, anogenital distance in female offspring, nipple retention and genital malformations in male offspring. The modeling was, however, not done for the latter endpoint due to lack of sufficient data for the single pesticides.

Data were normalized to the control groups of each year (experiment) so that year did not have to be included in the model as an extra factor. The effect of year was tested and either t test or lme in R was used.

The increase in gestation length was normalized to lie between zero and one, where the mean of the control group was the minimum (0) and the mean of the control group plus 2 was the maximum (1).

The endpoint nipple retention (NR) was scaled using the mean of the female controls (approximately 12–12.5). Values larger than this maximum were thresholded to the maximum. After scaling according to the female controls, the controls were not included in the analyses, but they are plotted in the figures as reference points.

The endpoint AGDI in female offspring was transformed for each experiment to lie between zero and one, scaled with the mean of the female controls as the minimum value, and the mean of the male controls as the maximum value.

For NR in males and AGDI in females, the same model and fitting method was used since these endpoints have several observations per litter. First, a binomial logit curve was fitted to the observations for each of the pesticides. Generalized Estimating Equations (GEEs) were used to perform the fit [35–37]. This method is particularly good when accounting for correlations within subject (here litter), but independence across litters, i.e. the model is based on pup level but taking the litter effect into account. The variance of the estimates can be incorrect if the correlation structure is not accounted for. In order to keep the number of parameters to a minimum in this model we assumed that the correlation is so called “exchangeable”, meaning that it is the same for any pair of litters.

The models were fitted using the function `gee` [38] in the R package `gee` from CRAN.

For gestation length, a generalized linear model with a binomial logit (GLM) was fitted to all observations for the five pesticides and the pesticide mixture. As gestation length was measured on dam level, there was only one observation per litter, and therefore a GLM rather than a GEE was used. The `glm()` function in the statistics program R was used for estimation and prediction.

Either a GEE or GLM was fitted to the same endpoints for both single pesticides and the mixture. Two models for prediction were considered. First, the dose-additivity predictions were estimated. These are defined by the effect dose of the pesticide mixture:

$$ED_{\text{mixture}} = \left(\frac{p_1}{ED_{X_1}} + \frac{p_2}{ED_{X_2}} + \frac{p_3}{ED_{X_3}} + \frac{p_4}{ED_{X_4}} + \frac{p_5}{ED_{X_5}} \right)^{-1}$$

for the effect doses of the five pesticides in question, and where p_i is the proportion of pesticide i in the mixture. The parametric delta method (closely related to parametric bootstrapping) was performed to estimate the ED_{mixture} from the fitted models and their variance estimates [39]. The distributions of the parameter estimates were sampled using the estimated variances of the parameters α and β . Random normal samples were used to sample new parameters and thereby obtain predictions using a Monte Carlo sampling strategy assuming a normal distribution of the parameters α and β . A thousand samples were drawn for each parameter, and a dense grid of the doses was used to estimate the effect doses.

Secondly, the independent actions model [33]:

$$E_{\text{mixture}} = 1 - (1 - E_{X_1}) \times (1 - E_{X_2}) \times (1 - E_{X_3}) \times (1 - E_{X_4}) \times (1 - E_{X_5})$$

was likewise estimated using such a Monte Carlo sampling. Here, E_{X_i} denotes the fractional effect (in %) of the i th pesticide, where we have $X_i = X_{\text{mixture}} \cdot p_i$. Note that the independent actions formula is given for the effect whereas the dose-additivity model is given in terms of the effect dose.

The same analyses were also performed with winsorization, as described by Scholze et al. [40]. This means that the fitting was performed iteratively with observations which fall outside 3 times the standard deviation of the mean prediction are adjusted to the value of 3 times the SD and then a new fit is performed etc.

2.8. In vitro assays

The pesticide mixture for the *in vitro* studies was designed based on the mixture applied for the *in vivo* studies. The ratio of the 5 pesticides (epoxiconazole, mancozeb, prochloraz, tebuconazole and procymidone) was:

	Epoxiconazole	Mancozeb	Prochloraz	Tebuconazole	Procymidone
mg basis	15	25	35	50	50
Molar basis	3	2	6	11	12

2.8.1. Androgen receptor reporter gene assay

The ability of the pesticides to activate the androgen receptor (AR) and to inhibit androgen-induced activation of the AR was tested in an AR reporter gene assay based on AR-transfected Chinese Hamster Ovary (CHO) cells as described by Kjaerstad et al. [41]. The test compounds were tested in triplicates within the range 0.03–30 μ M, and luciferase activity was measured as described by Kjaerstad et al. [41].

Cytotoxicity experiments were performed as described by Kjaerstad et al. [41].

2.8.2. T-screen for thyroid receptor activity

The T-screen assay was used for *in vitro* detection of the agonistic and antagonistic properties of the test compounds at the level of the thyroid hormone receptor (TR). The GH3 cells employed in the T-screen assay is a rat pituitary tumor cell line, which growth is dependent on the thyroid hormone (TH) 3,3',5-triiodothyronine (T3), and which has a high expression of TR. The growth stimulatory effect of T3 is mediated by specific, high-affinity TRs that upon binding of THs bind to thyroid hormone responsive elements (TREs) in the cell nucleus ultimately leading to gene expression and subsequent cell proliferation [42]. The rat pituitary cell line GH3 were obtained from American type culture collection (ATCC), and the assay was carried out as described by Taxvig et al. [43].

All compounds or mixtures were tested in triplicate (0, 0.01, 0.375, 1, 3, 10, and 30 μ M) and were tested both in the absence or presence of 0.22 nM T3 (T3-EC50) to test for agonistic and antagonistic potency. Control wells contained cells and test medium with the same amount of DMSO [0.1%] as the exposed cells. After incubation for 96 h, cell growth was measured using the dye resazurine [44]. 100 μ L of a 0.005 mg/ml resazurine solution in PBS, was added to each well, and plates were incubated 3 h at 37 °C, protected from light. Fluorescence was measured (excitation wavelength 560 nm/emission. 590 nm) on a Wallac Victor 1420 multilabel counter (PerkinElmer Life Sciences).

2.8.3. Steroid synthesis in the H295R cell line

The pesticides were investigated for effects on the synthesis of estradiol, progesterone, and testosterone in the human adrenocortical carcinoma cell line H295R (ATCC, CRL-2128) as described previously [45]. In brief, cells were seeded at a density of 3×10^5 cells/well in 24-well culture plates (Costar 3524, Corning, NY, USA) with DMEM/F12 medium (Gibco, Paisley, UK) supplemented with 2.0% Nu-serum (BD Sciences Denmark) along with 1% ITS + premix (containing 6.25 μ g/ml insulin, 6.25 μ g/ml transferrin, 6.25 ng/ml selenium, 1.25 mg/ml BSA and 5.35 μ g/ml lipoic acid; BDSciences Denmark) and incubated for 24 h at 37 °C in a humidified atmosphere of 5% CO₂/air.

The pesticides were added (1 ml) to the cells in triplicates at six concentrations. Control wells contained the same amount of DMSO (0.1%) as exposed cells. After 48 h incubation the medium was removed and stored at –20 °C until analysis. Hormones were measured using commercial hormone kits from Wallac Delfia (PerkinElmerDenmark, Hvidovre, Denmark). Cytotoxicity was evaluated by means of the MTT assay as described by Mosman [46], measuring the absorbance at 570 nm after 1.5–2 h incubation on a Wallac multilabel 1420 counter.

3. Results

3.1. Pregnancy, litter and offspring data

Pregnancy and litter data are shown in Table 2. There were no statistically significant effects on maternal body weight gain from GD 7 to GD 21 or from GD 7 to PD 1 in exposed dams compared to controls. No clinical signs of toxicity were observed in the dams and the number of implantation scars in the uterus, postimplantation and perinatal loss was similar among groups.

Gestation length was significantly increased in the two highest mixture groups (groups 3 and 4) compared to controls ($p < 0.01$ and $p < 0.001$, respectively) and in the group exposed to the highest dose of epoxiconazole ($p < 0.001$), but not in any other groups (Table 2).

Offspring data are shown in Table 3. Pup body weights were unaffected by pesticide exposure at birth and on PD 6, 13 and 22. Nipple retention in male offspring, an indication of anti-androgenic activity, was significantly increased in all mixture groups compared to controls ($p < 0.001$) (Fig. 1). There was also a significant increase

Table 2Pregnancy and litter data. Data represent group means based on litter means $\pm$ SD.

Dams and litters	1: Control	2: Pesti- mix-14.6	3: Pesti- mix-29.2	4: Pesti- mix-43.8	5: Epoxi-3.75	6: Epoxi-15	7: Manz-6.25	8: Manz-25	9: Prchl-8.75	10: Prchl-35	11: Tebu-12.5	12: Tebu-50	13: Procy-12.5	14: Procy-50
No. of dams (litters)	15	17	10 (9)	15 (14)	8	4	5	7	8	4	8	6	7	4
Maternal weight gain GD7–GD21	88.1 $\pm$ 13.2	88.6 $\pm$ 11.6	86.7 $\pm$ 8.6	77.1 $\pm$ 12.8	87.3 $\pm$ 16.1	84.0 $\pm$ 21.1	87.4 $\pm$ 5.7	84.1 $\pm$ 11.4	85.9 $\pm$ 9.6	85.0 $\pm$ 5.0	85.4 $\pm$ 15.6	82.5 $\pm$ 10.2	86.6 $\pm$ 10.0	85.0 $\pm$ 3.7
Maternal weight gain GD7–PD1	18.5 $\pm$ 9.8	17.4 $\pm$ 9.5	10.7 $\pm$ 8.3	14.6 $\pm$ 17.7	22.9 $\pm$ 10.2	20.8 $\pm$ 9.4	15.4 $\pm$ 7.3	13.3 $\pm$ 6.5	14.8 $\pm$ 4.8	8.8 $\pm$ 10.4	11.0 $\pm$ 9.6	17.3 $\pm$ 15.8	11.6 $\pm$ 8.2	6.3 $\pm$ 5.5
Body weight gain PD1–13	33.0 $\pm$ 9.5	35.5 $\pm$ 9.2	31.2 $\pm$ 9.3	22.9 $\pm$ 19.0	30.9 $\pm$ 9.7	33.5 $\pm$ 4.0	34.8 $\pm$ 8.6	37.3 $\pm$ 3.3	34.7 $\pm$ 6.1	36.0 $\pm$ 8.4	44.8 $\pm$ 9.2	33.7 $\pm$ 4.2	39.1 $\pm$ 6.5	41.8 $\pm$ 11.4
Gestation length (days)	23.0 $\pm$ 0.0	23.0 $\pm$ 0.0	23.5 $\pm$ 0.5**	23.6 $\pm$ 0.5*	23.1 $\pm$ 0.4	23.8 $\pm$ 0.5*	23.0 $\pm$ 0.0	23.0 $\pm$ 0.0	23.2 $\pm$ 0.4	22.8 $\pm$ 0.5	23.0 $\pm$ 0.0	23.1 $\pm$ 0.2	23.0 $\pm$ 0.0	23.0 $\pm$ 0.0
% Post-implantation loss	11.0 $\pm$ 24.9	5.4 $\pm$ 6.9	6.4 $\pm$ 9.9	16.7 $\pm$ 25.9	11.7 $\pm$ 11.7	4.0 $\pm$ 4.6	0.0 $\pm$ 0.0	2.3 $\pm$ 4.0	3.7 $\pm$ 6.0	1.8 $\pm$ 3.6	7.9 $\pm$ 10.5	5.5 $\pm$ 6.5	13.4 $\pm$ 35.1	0.0 $\pm$ 0.0
% Perinatal loss	14.9 $\pm$ 25.2	7.3 $\pm$ 8.6	23.4 $\pm$ 28.3	23.3 $\pm$ 28.6	13.2 $\pm$ 12.6	10.3 $\pm$ 15.8	1.8 $\pm$ 4.1	3.3 $\pm$ 4.2	5.4 $\pm$ 7.0	1.8 $\pm$ 3.6	8.6 $\pm$ 12.3	9.5 $\pm$ 5.7	17.3 $\pm$ 36.9	1.9 $\pm$ 3.8
Born alive per. litter	11.1 $\pm$ 2.3	11.9 $\pm$ 2.5	11.6 $\pm$ 2.4	9.8 $\pm$ 2.0	10.4 $\pm$ 1.8	8.8 $\pm$ 3.7	12.2 $\pm$ 1.2	12.9 $\pm$ 1.7	11.7 $\pm$ 1.8	12.0 $\pm$ 1.7	11.8 $\pm$ 1.9	11.5 $\pm$ 2.5	10.8 $\pm$ 3.0	12.8 $\pm$ 2.4
% Postnatal death	4.4 $\pm$ 9.4	2.1 $\pm$ 4.3	17.4 $\pm$ 30.8	8.9 $\pm$ 18.1	1.8 $\pm$ 5.1	6.8 $\pm$ 13.6	1.8 $\pm$ 4.1	1.0 $\pm$ 2.7	1.7 $\pm$ 5.1	0.0 $\pm$ 0.0	1.0 $\pm$ 2.9	4.1 $\pm$ 6.6	2.0 $\pm$ 5.4	1.9 $\pm$ 3.9
% Males	48.4 $\pm$ 18.8	47.4 $\pm$ 17.5	52.4 $\pm$ 13.5	51.6 $\pm$ 16.3	44.9 $\pm$ 28.1	36.0 $\pm$ 29.6	49.8 $\pm$ 13.7	45.4 $\pm$ 12.9	40.2 $\pm$ 11.9	54.3 $\pm$ 12.7	56.2 $\pm$ 15.2	47.4 $\pm$ 16.8	44.6 $\pm$ 19.6	53.6 $\pm$ 17.3

Epoxi: epoxiconazole; Manz: mancozeb; Prchl: prochloraz; Tebu: tebuconazole; Procy: procymidon; Perinatal loss: loss from implantation to weaning; Postnatal death: death after birth.

* $p < 0.05$.** $p < 0.01$.**Table 3**Offspring data. Data represent group means based on litter means $\pm$ SD. Mix = mixture.

Offspring (data from viable litters)	1: Control	2: pestimix-14.6	3: pesti-mix-29.2	4: Pestimix-43.8	5: Epoxi3.75	6: Epoxi15	7: Manz-6.25	8: Manz-25	9: Prchl-8.75	10: Prchl-35	11: Tebu-12.5	12: Tebu-50	13: Procy-12.5	14: Procy-50
Male birth weight (g)	6.5 $\pm$ 0.4	6.1 $\pm$ 0.4	6.2 $\pm$ 0.4	6.4 $\pm$ 0.5	6.5 $\pm$ 0.2	6.6 $\pm$ 0.6	6.2 $\pm$ 0.1	6.2 $\pm$ 0.2	6.4 $\pm$ 0.4	6.3 $\pm$ 0.4	6.3 $\pm$ 0.4	6.1 $\pm$ 0.4	6.2 $\pm$ 0.3	6.4 $\pm$ 0.3
Female birth weight (g)	6.1 $\pm$ 0.4	5.9 $\pm$ 0.4	5.9 $\pm$ 0.5	6.1 $\pm$ 0.6	6.1 $\pm$ 0.3	6.3 $\pm$ 0.6	6.1 $\pm$ 0.2	5.9 $\pm$ 0.1	6.0 $\pm$ 0.4	6.0 $\pm$ 0.4	6.0 $\pm$ 0.3	5.9 $\pm$ 0.2	6.1 $\pm$ 0.2	6.1 $\pm$ 0.3
Male AGD (units)	24.5 $\pm$ 1.1	23.4 $\pm$ 1.2	24.1 $\pm$ 1.0	24.0 $\pm$ 1.2	25.7 $\pm$ 0.9***	23.9 $\pm$ 1.3	23.9 $\pm$ 1.0	24.0 $\pm$ 0.9	24.8 $\pm$ 0.6	24.6 $\pm$ 0.7	24.7 $\pm$ 1.3	24.6 $\pm$ 0.8	22.8 $\pm$ 0.9**	22.0 $\pm$ 0.7***
Male AGD/cubic root bw (AGDI)	13.1 $\pm$ 0.5	12.8 $\pm$ 0.7	13.1 $\pm$ 0.5	12.9 $\pm$ 0.4	13.8 $\pm$ 0.4**	12.7 $\pm$ 1.0	13.0 $\pm$ 0.6	13.1 $\pm$ 0.4	13.4 $\pm$ 0.4	13.4 $\pm$ 0.4	13.4 $\pm$ 0.7	13.5 $\pm$ 0.4	12.4 $\pm$ 0.4**	11.9 $\pm$ 0.3***
Female AGD (units)	13.6 $\pm$ 0.6	13.7 $\pm$ 0.8	14.1 $\pm$ 0.5	15.0 $\pm$ 0.8***	14.9 $\pm$ 1.0**	14.3 $\pm$ 0.9	14.2 $\pm$ 0.9	13.5 $\pm$ 0.5	14.1 $\pm$ 0.8	14.5 $\pm$ 0.4*	14.4 $\pm$ 1.0	14.5 $\pm$ 0.5*	13.4 $\pm$ 0.4	13.3 $\pm$ 0.2
Fem. AGD/cubic root bw (AGDI)	7.4 $\pm$ 0.3	7.6 $\pm$ 0.5	7.8 $\pm$ 0.3*	8.2 $\pm$ 0.3***	8.1 $\pm$ 0.6**	7.8 $\pm$ 0.7	7.8 $\pm$ 0.5	7.5 $\pm$ 0.2	7.8 $\pm$ 0.5*	8.0 $\pm$ 0.2**	7.9 $\pm$ 0.6*	8.0 $\pm$ 0.2**	7.3 $\pm$ 0.2	7.3 $\pm$ 0.2
No. Areolas males ^a	0.0 $\pm$ 0.0	1.9 $\pm$ 1.6***	3.3 $\pm$ 0.9***	5.3 $\pm$ 1.1***	0.1 $\pm$ 0.2	0.5 $\pm$ 1.0	0.1 $\pm$ 0.1	0.6 $\pm$ 0.6*	0.3 $\pm$ 0.2	1.7 $\pm$ 1.2***	0.5 $\pm$ 0.8	1.6 $\pm$ 0.4***	2.8 $\pm$ 1.2***	6.0 $\pm$ 2.0***
No. Areolas. females	12.0 $\pm$ 0.4	13.4 $\pm$ 3.9	12.4 $\pm$ 0.4	12.5 $\pm$ 0.4	12.4 $\pm$ 0.3	12.8 $\pm$ 0.5	12.2 $\pm$ 0.1	12.4 $\pm$ 0.1	12.3 $\pm$ 0.2	12.1 $\pm$ 0.2	12.5 $\pm$ 0.6	12.2 $\pm$ 0.2	12.2 $\pm$ 0.2	12.4 $\pm$ 0.4
Male body weight PD 6 (g)	12.9 $\pm$ 1.3	12.2 $\pm$ 1.3	12.0 $\pm$ 1.3	12.5 $\pm$ 1.5	13.1 $\pm$ 0.9	13.2 $\pm$ 1.0	12.3 $\pm$ 0.7	11.8 $\pm$ 0.4	12.7 $\pm$ 1.2	12.9 $\pm$ 1.7	12.1 $\pm$ 1.7	12.3 $\pm$ 1.2	12.4 $\pm$ 0.7	12.2 $\pm$ 1.4
Female body weight PD 6 (g)	12.5 $\pm$ 1.2	11.9 $\pm$ 1.4	11.6 $\pm$ 1.4	11.9 $\pm$ 1.3	12.8 $\pm$ 1.4	12.2 $\pm$ 1.0	12.1 $\pm$ 0.6	11.4 $\pm$ 0.5	12.1 $\pm$ 1.2	12.2 $\pm$ 1.7	11.9 $\pm$ 1.7	12.1 $\pm$ 1.2	12.2 $\pm$ 0.7	12.0 $\pm$ 1.1
Male body weight PD 13 (g)	26.1 $\pm$ 4.1	24.0 $\pm$ 2.7	25.1 $\pm$ 2.9	26.7 $\pm$ 3.9	25.9 $\pm$ 3.0	28.0 $\pm$ 2.4	24.6 $\pm$ 2.4	23.0 $\pm$ 1.0	25.6 $\pm$ 1.5	24.9 $\pm$ 2.7	24.2 $\pm$ 2.7	24.8 $\pm$ 2.0	24.9 $\pm$ 1.0	24.3 $\pm$ 2.9
Fem. body weight PD 13 (g)	25.4 $\pm$ 3.9	23.7 $\pm$ 3.0	24.4 $\pm$ 2.8	24.3 $\pm$ 4.6	26.0 $\pm$ 4.1	26.5 $\pm$ 2.2	24.4 $\pm$ 2.5	22.7 $\pm$ 1.4	24.6 $\pm$ 1.8	24.0 $\pm$ 2.7	23.9 $\pm$ 2.1	24.4 $\pm$ 2.4	24.5 $\pm$ 1.4	23.6 $\pm$ 2.8
Male body weight PD 22 (g)	45.6 $\pm$ 6.7	41.5 $\pm$ 4.4	44.0 $\pm$ 5.6	46.8 $\pm$ 7.0	43.3 $\pm$ 4.6	46.8 $\pm$ 3.5	42.9 $\pm$ 4.2	40.4 $\pm$ 1.6	43.2 $\pm$ 3.3	42.4 $\pm$ 5.0	43.0 $\pm$ 4.6	43.8 $\pm$ 4.5	42.5 $\pm$ 1.3	43.0 $\pm$ 4.4
Fem. body weight PD 22 (g)	45.2 $\pm$ 6.5	41.1 $\pm$ 4.2	41.9 $\pm$ 3.9	44.8 $\pm$ 5.8	46.3 $\pm$ 7.7	46.7 $\pm$ 3.2	42.6 $\pm$ 3.5	40.3 $\pm$ 2.2	42.3 $\pm$ 4.0	41.3 $\pm$ 4.9	42.7 $\pm$ 3.4	42.8 $\pm$ 4.5	42.4 $\pm$ 1.3	41.3 $\pm$ 4.9

Epoxi: epoxiconazole; Manz: mancozeb; Prchl: prochloraz; Tebu: tebuconazole; Procy: procymidon.

^a The number of nipples in the control group males was zero and it was therefore necessary to put in 1 nipple in 3 pups from the control group to perform the statistical model.* $p < 0.05$.** $p < 0.01$.*** $p < 0.001$.

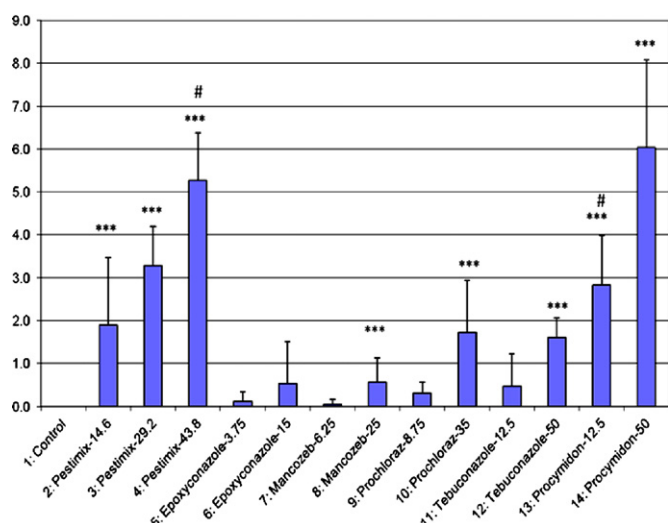


Fig. 1. Effects of single and combined exposure to procymidone, prochloraz, tebuconazole, epoxiconazole and mancozeb on nipple retention in male pups PD 13. Results shown are group means $\pm$ standard deviation. See Table 3 for more details. The number of nipples in female controls is normally 12. ****p*-Values are <0.001 . #The difference between Pestimix-43.8 (group 4) and procymidone-12.5 (group 13) is statistically significant ($p < 0.001$).

in the groups exposed to the high dose of mancozeb ($p < 0.05$), prochloraz ($p < 0.001$), tebuconazole ($p < 0.001$) and both doses of procymidone ($p < 0.001$) (Fig. 1). The magnitude of the effects in the highest mixture group was significantly higher than the effect induced by the low dose of procymidone ($p < 0.001$), although the procymidone dose was the same in the two groups.

In the male offspring, no statistically significant changes on anogenital distance (AGD) were seen in the mixture groups. The data for the single chemical exposures showed that the low dose of epoxiconazole induced longer AGD in male pups compared to controls ($p < 0.01$), whereas the high dose had no effect. Both doses of procymidone induced shorter male AGD ($p < 0.01$, and $p < 0.001$, respectively). In the female offspring, a dose-dependent increase in anogenital distance was seen, with significantly longer AGDI, in the two highest mixture groups compared to controls ($p < 0.05$ and $p < 0.001$, respectively). A similar effect was seen in animals exposed to both doses of prochloraz ($p < 0.05$ and $p < 0.01$) and tebuconazole ($p < 0.05$ and $p = 0.001$) and in the group exposed to the low dose of epoxiconazole ($p < 0.01$) but not in female pups exposed to a four times higher dose of epoxiconazole (Table 3).

3.2. Genital malformations in male offspring

The results of the scoring of external genital malformations in male offspring sacrificed on PD 16 and 22 indicated increased number and severity of genital malformation in the highest mixture group and the highest dose of procymidone, but the differences were not statistically significant compared to controls shown in Supplemental Material Tables X2). No animals had increased gubernacular length as an indication of cryptorchidism.

The rest of the male animals were scored for external genital malformations alive shortly after sexual maturation (around PD 50) and the results were combined for all time points (see Fig. 2 and Table 4). The combined results showed significantly increased number and severity of external genital malformation in the highest mixture group ($p = 0.006$) and the highest dose of procymidone ($p = 0.035$), compared to controls. The group mean of the percentage of animals within each litter with external genital malformations was markedly higher in the highest mixture group than in any of the groups dosed with the individual pesticides alone at the dose

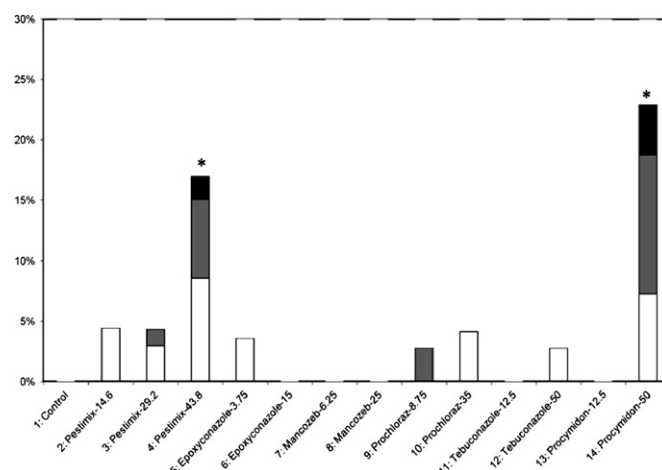


Fig. 2. Genital malformations in male offspring. Results shown are the group mean of the % of the male offspring within each litter given score 1 (white), score 2 (gray) or score 3 (black). The rest of the animals, i.e. up to 100%, within each group were given score 0. The incidence of genital malformations is significantly increased ($p = 0.017$) in the highest mixture group (Pestimix-43.8, group 4) and the high dose of procymidone (group 14) ($p = 0.035$).

included in the mixture (around 17% compared to 0–4%, Fig. 2). The individual pesticides alone caused no statistically significant effect at the dose levels included in the mixture.

External malformations related to the reproductive system and seminal vesicles were observed in a few of the nine months old animals (Supplemental Material Table X3). Cryptorchidism and alopecia, in the perineal area, were not observed in any of the adult males. Two males in the highest mixture group had penile malformation: one had both a severe cleft penis and hypospadias with urethral opening at the basis of the penis (score 3), the other had a cleft phallus (score 1). A third male in the highest mixture group had a malformed seminal vesicle. One male in each of the groups dosed with mancozeb showed malformation of the seminal vesicle. In the group dosed with the highest level of procymidone two animals had cleft phallus (scores 2 and 3, respectively), and a third animal had malformed seminal vesicle. However, none of the observed effects were significantly different from control animals.

3.3. Mixture modeling – gestation length

General linear model (GLM) fits for the five pesticides are illustrated in Supplemental Material Fig. X1. The effect on gestation length of mancozeb and procymidone was not statistically significant ($p = 0.25$ for both pesticides). Therefore, mancozeb and procymidone were not included in the predictions for the mixture. A fit of the observed pesticide mixture was performed similar to the single pesticide observations. In addition, predictions were performed based on the models for the single pesticides and the dose-additivity and independent addition formulas. The results are shown in Table 5 and Fig. 3A and B. At high doses, there was a significant difference between the observed and predicted mixture effects, and both of the models underestimated the mixture effects. However, the underestimation was generally smaller for the dose-additivity prediction than the independent action prediction. The dose-addition prediction at the lower doses gave a good prediction of the observed effects. In Supplemental Material Table X4 the effect doses for increased gestation length for individual pesticides are summarized. The overall results were the same when winsorization was used (data not shown).

Table 4

External genital malformations in male pups scored on PD 16 or 22 at sectioning or *in vivo* on PD 50. Values show the number of animals with score 0 (normal), score 1 (slight), score 2 (moderate), or score 3 (severe).

Dose group	No. animals (litters)	Score 0	Score 1	Score 2	Score 3	Score 1, 2 or 3	Statistics
1: Control	71 (15)	71 (15)					
2: Pestimix-14.6	96 (17)	92 (17)	4 (4)			4 (4)	NS
3: Pestimix-29.2	53 (9)	50 (9)	2 (2)	1 (1)		3 (2)	NS
4: Pestimix-43.8	67 (14)	56 (14)	4 (3)	5 (4)	2 (2)	11 (6)*	$p = 0.006$
5: Epoxi-3.75	38 (7)	36 (7)	2 (1)				
6: Epoxi-15	11 (4)	11 (4)					
7: Manz- 6.25	31 (5)	31 (5)					
8: Manz-25	39 (7)	39 (7)					
9: Prchl-8.75	44 (9)	43 (9)	0	1 (1)		1 (1)	NS
10: Prchl-35	26 (4)	25 (4)	1 (1)			1 (1)	NS
11: Tebu-12.5	48 (8)	48 (8)					
12: Tebu-50	27 (6)	26 (6)	1 (1)			1 (1)	NS
13: Procy-12.5	38 (7)	38 (7)					
14: Procy-50	25 (4)	19 (4)	2 (2)	3 (2)	1 (1)	6 (2)*	$p = 0.035$

Values in parentheses are the number of litters with offspring having the indicated score. Epoxi: epoxiconazole; manz: mancozeb; Prchl: prochloraz; Tebu: tebuconazole; Procy: procymidon.

* $p < 0.05$.

Table 5

Observed and predicted mixture effect doses for increased gestation length (GL). Results shown are mean dose [95% confidence interval].

Relative GL GL (rounded)	10% 23.2	25% 23.5	50% 24.0	75% 24.5	90% 24.8
Observed ^a	18.0 [3.4–31.0]	46.5 [37.5–57.5]	75.5 [64.5–93.5]	104.0 [85.0–127.0]	133.0 [101.5–159.0]
Dose-additivity	9.49 [8.50–10.5]	64.0 [61.6–66.5]	135.3 [132.8–137.9]	197.8 [195.1–200.5]	256.2 [253.2–259.3]
Independent action	0.5 [0.14–0.86]	0.5 [0–4.05]	173.0 [167.4–178.6]	316.5 [310.5–322.5]	439.5 [433.0–446.0]

^a Based on the curve fit shown in Fig. 3A.

3.4. Mixture modeling – nipple retention in male offspring

In [Supplemental Material Fig. X1](#) the nipple retention results for the single chemicals are shown with the estimated model and ± 2 standard errors. No statistically significant effect on NR by mancozeb was observed and the effect was therefore set to zero in the predictions.

The fitted model of the observed results and the dose-additivity predictions are illustrated in [Fig. 4A](#) and [B](#). Furthermore, the 95% confidence intervals and the estimates of observed data, the dose-additivity predictions, and the independent actions predictions are summarized in [Table 6](#). The results show that for the low doses, the confidence intervals overlapped and there were no apparent statistical differences between the observed dose effects and the dose-additivity predictions. However, at the high doses no overlap in the confidence intervals between the observed dose effects and those predicted by dose-additivity was found. In fact, the dose-additivity formula underestimated the effects for the high doses of the mixture. The independent action predictions were seen to likewise underestimate the effects for high doses, and overestimated the effect at low doses. Furthermore, the independent actions model gave poorer estimates in relation to the observed effects, in comparison to the dose-additivity model. In [Supplemental Material Table X5](#) the effect doses for increased nipple retention in male offspring for individual pesticides are summarized. By using winsorization similar results were obtained (data not shown).

3.5. Mixture modeling – AGDI in female offspring

As there was a significant effect of year on AGDI in female offspring, transformed AGDI observations were used in the analysis. The same model and fitting method (GEE) as for NR was used for effects on AGDI. However, only prochloraz could, based on this, be assumed to have a statistically significant effect on AGDI (data not shown). The relatively few data available for epoxiconazole and tebuconazole suggest that accepting the null-hypothesis

of no effect from these pesticides without further examination is arguable.

The fitted model of the observed results is illustrated in [Fig. 5A](#). The results in [Table 7](#) and [Fig. 5B](#) show that both independent action and dose-additivity predictions highly underestimated the effect of the pesticide mixture. So, although no statistical effect was seen for epoxiconazole, mancozeb, tebuconazole and procymidone individually, a marked contribution to the effect of the mixture was evident. In [Supplemental Material Table X6](#), the effect doses for longer AGDI in female offspring for prochloraz are shown. Using winsorization, the estimated effect doses were slightly higher for the low effect doses and slightly lower for the high effect doses (data not shown). However, the overall picture was the same.

3.6. In vitro studies

All pesticides except for mancozeb exhibited AR antagonism *in vitro* ([Fig. 6](#)). The lowest observed effect levels (LOEC) and cytotoxic concentrations are shown in [Supplemental Material Table X7](#). The ranking of potencies for AR antagonism was: mixture $\sim$ procymidone $>$ prochloraz $\sim$ epoxiconazole $>$ tebuconazole. As the content of procymidone in the mixture constitutes only approximately 1/3 of the total mixture dose, these results indicate combination effects on AR antagonism *in vitro*.

In the H295R steroid synthesis assay, epoxiconazole, prochloraz, tebuconazole, and the mixture were shown to reduce testosterone levels in the cells, whereas mancozeb and procymidone had no effect ([Fig. 7](#)). Epoxiconazole, prochloraz, tebuconazole and the mixture also reduced estradiol levels ([Fig. 7](#)). Generally the pesticides and the mixture caused an increase in progesterone levels, although for the mixture and for procymidone the effect was only seen at the highest concentrations tested ([Fig. 7](#)). In general the effects of prochloraz was most pronounced on all three hormones.

Triiodothyronine (T_3) induced dose-dependent proliferation of the GH3 cells in the T-screen assay as expected (data not shown). No agonistic effect was observed for the mixture ([Supplemental](#)

Table 6
Observed and predicted mixture effect doses for the increased nipple retention (NR) in male offspring. Results shown are mean dose [95% confidence interval].

Relative NR NR (rounded)	10% 1	25% 3	50% 6	75% 9	90% 11
Observed ^a	2.3 [0.01–4.9]	27.0 [25.5–28.5]	51.5 [49.5–53.5]	76.0 [72.5–79.5]	100.5 [95.0–105.5]
Dose-additivity	0.021 [0.020–0.022]	18.85 [17.6–20.1]	98.39 [97.3–99.5]	153.53 [151.9–155.1]	206.45 [204.3–208.5]
Independent action	0.01 [0.01–0.01]	0.01 [0.01–0.01]	41.5 [39.8–43.1]	184.0 [182.5–185.5]	289.0 [287.4–290.6]

^a Based on the curve fit shown in Fig. 4A.

Table 7
Observed and predicted mixture effect doses for increased anogenital index (AGDI) in female offspring. Results shown are mean dose [95% confidence interval].

Relative AGDI AGDI (rounded)	10% 8.04	25% 8.89	50% 10.30	75% 11.72	90% 12.56
Observed ^a	38.0 [34.0–42.0]	104.0 [92.0–124.0]	170.0 [147.0–202.5]	236.0 [199.0–275.0]	302.0 [245.0–346.5]
Dose-additivity	224.0 [217.8–230.3]	720.6 [709.6–731.6]	1217 [1200–1234]	1714 [1691–1737]	2210 [2180–2239]
Independent action	209.5 [202.9–216.1]	694.5 [687.9–701.1]	1228 [1211–1245]	1750 [1726–1773]	2255 [2226–2285]

^a Based on the curve fit shown in Fig. 5A.

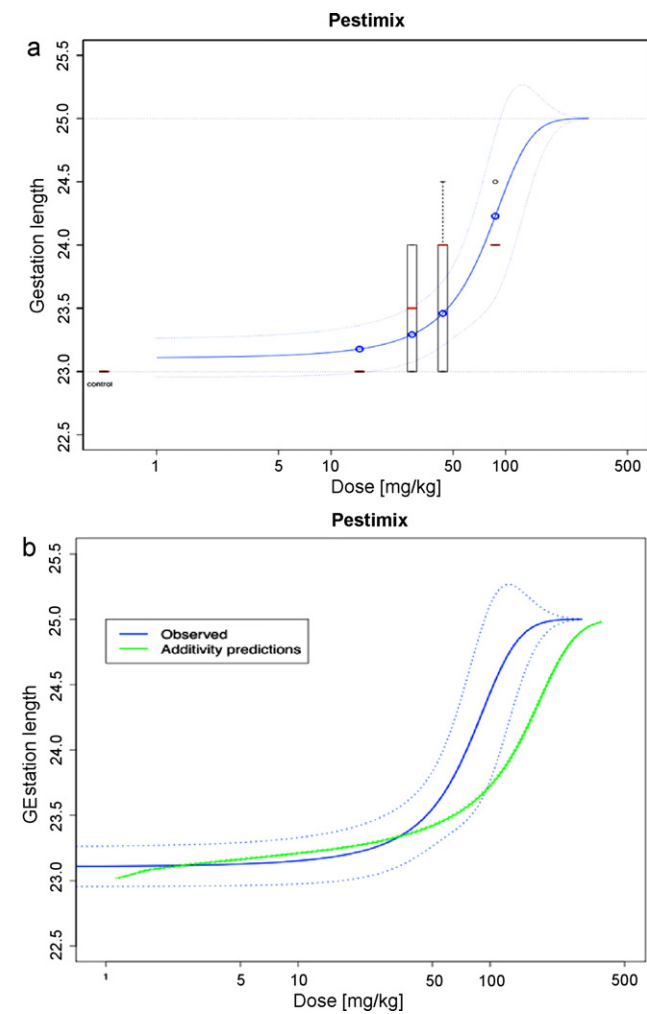


Fig. 3. (A) GLM fit of mixture effect on gestation length (GL). The observations are shown as box plots, red marks=median of observations at a given dose. (B) The observed and predicted mixture effects on gestation length GL. The dose-additivity prediction of the mixture effect doses are in green and the observed curve is in blue. In dotted lines ± 2 standard errors are illustrated. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of the article.)

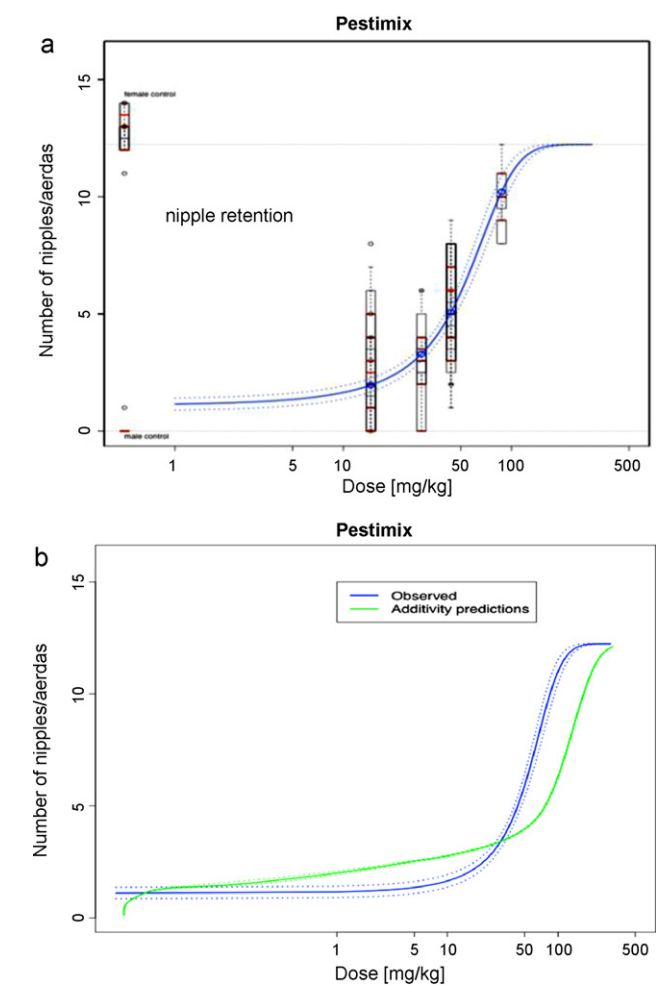


Fig. 4. (A) GEE fit of mixture effects on nipple retention in male offspring ± 2 standard errors are illustrated with dashed lines (in blue). The observations are shown as box plots of each litter with red marks at the litter medians. (B) The observed and predicted mixture effect on nipple retention in male offspring. The dose-additivity predictions of the mixture effect doses are in green and the observed curve is in blue. In dotted lines ± 2 standard errors are illustrated. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of the article.)

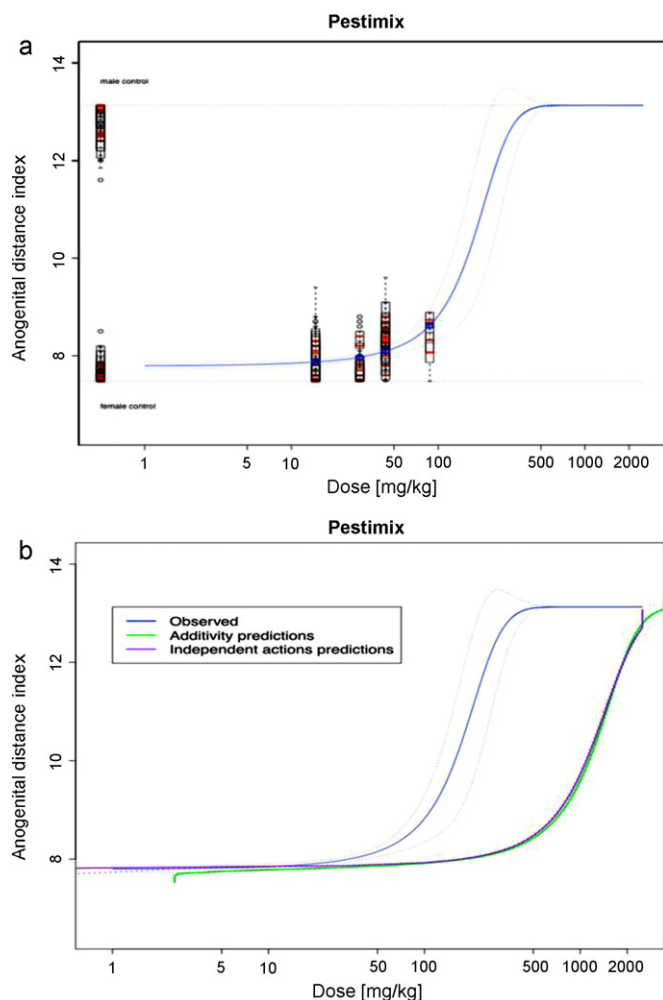


Fig. 5. (A) GEE fit of mixture effects on anogenital index (AGDI) in female offspring ± 2 standard errors are illustrated with dashed lines (in blue). The observations are shown as box plots of each litter with red marks at the litter medians. (B) The observed and predicted mixture doses for effects on anogenital index in female offspring. The dose-additivity predictions are in green, the independent action predictions are in purple. And the observed curve in blue. In dotted lines ± 2 standard errors are illustrated. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of the article.)

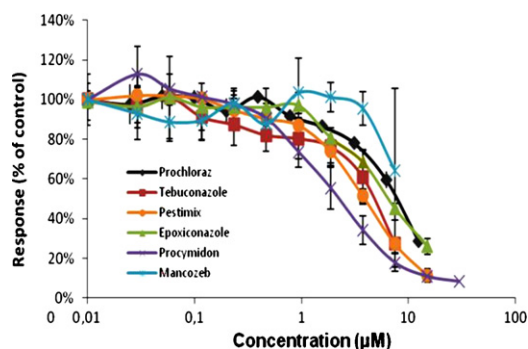


Fig. 6. Androgen receptor antagonism of single pesticides and the mixture (pestimix) *in vitro*.

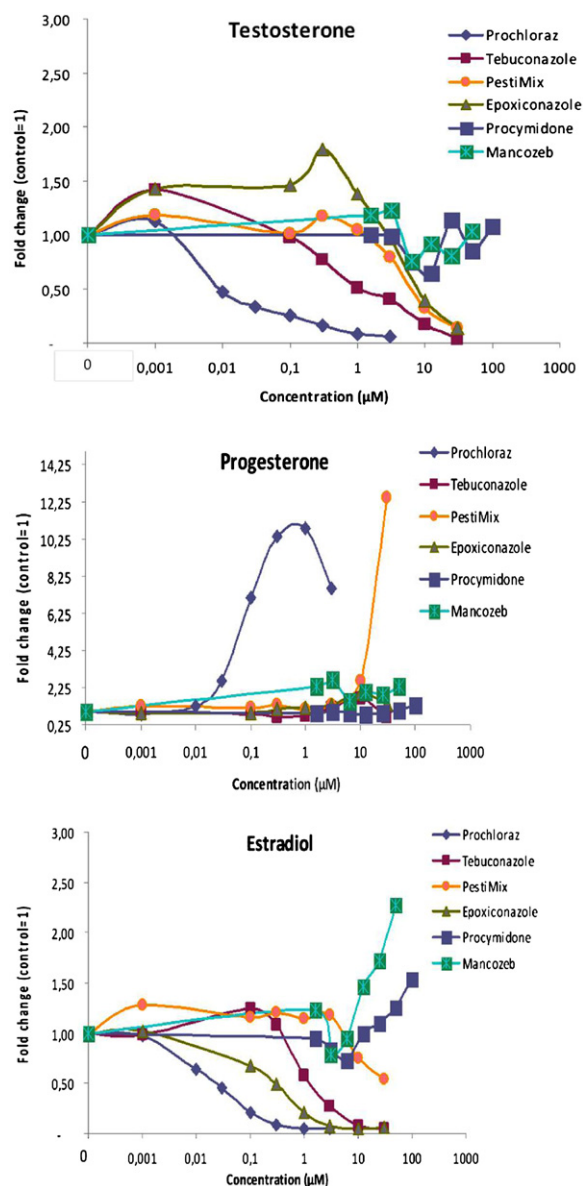


Fig. 7. Changes in levels of testosterone (top) estradiol (middle) and progesterone (bottom) in the H295 steroid synthesis assay after exposure to increasing concentrations of the 5 pesticides singly or in combination.

Material Fig. X3a). A weak antagonistic effect of the mixture was observed at concentrations at and above $3.13 \mu\text{M}$ (**Supplemental Material Fig. X3b**). A summary of all *in vitro* results is shown in **Supplemental Material Table X7**.

4. Discussion

This study generally aimed to explore the hypothesis that combined developmental exposure to endocrine disrupting pesticides could lead to adverse developmental toxicity effects at dose levels below NOAELs for the single pesticides, and to investigate if dose-additivity modeling of the expected mixture effects could give useful estimates of the observed mixture effects.

4.1. Developmental mixture study and mixture modeling

There were no statistically significant effects of pesticide exposure on maternal body weight gain or on pup body weights at birth or in the postnatal period. For the mixture, this was consistent

with the results from a previous range-finding study [23] in which maternal and pup body weight gains were only affected at much higher doses of the mixture.

Several of the investigated endocrine and reproductive endpoints were affected by the combination of the five pesticides, and the effects were seen at dose levels at which the individual pesticides caused no, or only minor, effects (Table 8). Gestation length was significantly increased in the two highest mixture groups and in the group exposed to the highest dose of epoxiconazole, but not in any other group. The mixture doses consisted of each of the pesticides at 17% or 25% of their NOAEL for effect on gestation length based on results of previous studies in our laboratory. As the highest dose of epoxiconazole given alone was four times higher than the dose included in the highest mixture group, these results showed a combination effect of the pesticides at dose levels where the individual pesticides caused no effects. Increased gestation length of very similar magnitude was also observed in a similarly exposed mixture group in the range-finding study [23]. In that study, combined exposure at higher doses induced severe effects manifested as dystochia (impaired parturition) and high perinatal pup mortality [23].

To our knowledge, this is the first study to describe mixture effects on gestation length. The mixture effect is probably due to the presence of the three azole fungicides in the mixture, epoxiconazole, prochloraz and tebuconazole, which have previously been shown to elicit such effects on gestation length [18,47,48]. Prolongation of the gestation period may possibly be due to an increase in progesterone in the dams as suggested for epoxiconazole, prochloraz and tebuconazole [18,20]. It has previously been shown that all three azoles are able to induce progesterone levels *in vitro* [41] and this may be the cause of the increased gestation length *in vivo* as a decline in progesterone levels in the dams is a prerequisite for on-time delivery in rats [49].

The dystochia and pup mortality seen in the range finding studies have previously been observed as common effects for several azole fungicides [18,47,50,51]. The mode of action for these endpoints may be increased progesterone levels as well, and if this is the case, then these azole fungicides should be considered as similarly acting. However, further data are needed to make firm conclusions. Neither mancozeb nor procymidone have previously been shown to affect gestation length or perinatal survival at the doses studied [7,21].

For the endpoint gestation length, application of the dose-addition model at low doses resulted in a good prediction of the observed mixture effects. Both dose-additivity and independent action model predictions underestimated the mixture effects at high doses. However, the underestimation was generally smaller for the dose-additivity prediction than the independent action prediction. Our hypothesis is that the underestimation at high doses may be due to toxicokinetic interactions as e.g. saturation of metabolism. We have preliminary data to support this view as higher pup blood levels of some of the pesticides was found after mixture exposure than after exposure to the pesticides alone [25]. This is the first study to show mixture effects on the complex endpoint gestation length, and also the first study to show that it is possible to predict mixture effects using dose-additivity for this more 'functional' and biologically relevant endpoint as well.

Nipple retention in male offspring was significantly increased in all mixture groups. For the highest mixture group, the results were consistent with the results from the range-finding study [23]. Nipple retention was also increased at the highest dose of mancozeb, prochloraz and tebuconazole and at both doses of procymidone. However, the effect of the high dose of mancozeb (25 mg/kg) was numerically quite small. As an earlier study of mancozeb including larger groups and higher dose levels (up to 100 mg/kg) did not show any effect on nipple retention [21], the significant effect in

Table 8
Summary of effects found in dams and male rat pups exposed to epoxiconazole (Epoxi), mancozeb (Manz), prochloraz (Prchl), tebuconazole (Tebu) or procymidone (Procy), or a mixture of these pesticides (Pestimix) from GD 7 to PD 16.

Endpoint	Epoxi 3.75 mg/kg	Manz 6.25 mg/kg	Prchl 8.75 mg/kg	Tebu 12.5 mg/kg	PROcy 12.5 mg/kg	Pestimix 43.8 mg/kg	Dose-additivity ?	Joint effect compared to effect of single chemicals at Mix-25%
Gestation length, days	23.1	23.0	23.2	23.0	23.0	23.6	Yes, slight synergy at high doses	Marked joint effect; no significant effect of single chemicals
Nipple retention, PND 13, number	0.1	0.1	0.3	0.5	2.8	5.3	Yes, slight synergy at high doses	Marked joint effect; smaller significant effect of procymidone
Genital malformations, %								
Score 1	3.6	0.0	0.0	0.0	0.0	8.6	n.a.	Marked joint effect; no significant effect of single chemicals
Score 2	0.0	0.0	2.8	0.0	0.0	6.5		
Score 3	0.0	0.0	0.0	0.0	0.0	1.9		
Total	3.6	0.0	2.8	0.0	0.0	17.0		

n.a.: not analyzed. Values are expressed as percent of control values for relative organ weights and as increase in number for nipple retention (control value = 0.0). Values for genital malformations of external male reproductive organs are shown as group mean of % offspring within each litter having score 1, 2 or 3 and the sum (control value = 0.0%). Statistically significant effects compared to controls are indicated by bold letters.

the present study is likely to be a random finding. For the other pesticides, similar effects have to some extent been seen for the individual pesticides [6,16,18,20,26,47,51–53]. No statistically significant effects on nipple retention of mancozeb, prochloraz and tebuconazole were found at the low dose level, whereas procymidone caused a significant effect (Table 8). Results on nipple retention indicate mixture effects, since the magnitude of the effect in the highest mixture group was significantly higher than the effect induced by procymidone alone at the dose level included in the highest mixture. Mixture effects for nipple retention have earlier been shown after exposure to other mixtures of both similarly and dissimilarly acting anti-androgens in several rat studies [6,26,54].

The mixture effect prediction based on dose-additivity was in agreement with the observed effects at low doses, as there were no statistically significant differences between the predicted and the observed effect doses. However, dose-additivity generally underestimated the effects for the high doses of the mixture. The independent action predictions resulted in even stronger underestimations of the effects for high doses, and overestimations of the effect for the low doses. It has been reported previously that dose-additivity gives a good prediction of mixture effects of nipple retention with both similarly and dissimilarly acting anti-androgens [6,26,54].

In male offspring, no statistically significant effects on anogenital distance (AGD) were seen in the mixture groups. At the dose levels included in the mixture, epoxiconazole induced longer AGD, whereas procymidone induced shorter AGD. Consequently, the lack of effect in the mixture group might be due to opposite actions of these compounds.

In the female offspring, significantly longer AGDI, was seen at the two highest mixture doses. A similar effect was seen in female offspring given the low dose levels of epoxiconazole, prochloraz and tebuconazole, i.e. at doses equal to those included in the highest mixture dose. No effect was seen in female offspring dosed with a four times higher dose of epoxiconazole and we hypothesize that the finding at the low dose may be a random finding or alternatively, the lack of effect at the high dose may be due to a limited group size. However, the high doses of both prochloraz and tebuconazole did cause increased female AGDI in the present study. The AGD effect caused by the mixture is probably due to the three azole fungicides as earlier studies in our laboratory have shown similar effects for epoxiconazole, prochloraz and tebuconazole [18,53]. The mechanism underlying the increased AGDI in the females may be the increased progesterone levels in the dams causing virilization of the female genitals [55].

Both independent action and dose-additivity predictions highly underestimated the effect of the pesticide mixture on AGDI in females. So, although statistically significant effect was neither seen in the combined analysis of the results in the present study, nor in the historical data for epoxiconazole, mancozeb, tebuconazole and procymidone individually, some of the pesticides seem to have contributed to the effect observed after mixture exposure. The present study showed significant effects of the low doses of epoxiconazole, prochloraz and tebuconazole, whereas no effect on female AGDI was seen for mancozeb and procymidone. Earlier studies in our laboratory have also shown similar effects of epoxiconazole, prochloraz and tebuconazole [18,53]. The effect in the mixture groups was therefore most likely caused by the combined exposure to the three azole fungicides. The finding that mixed exposure to endocrine disrupters can lead to increased female AGD is new and more knowledge on this endpoint is needed.

An increased incidence and severity of malformations in the highest mixture group and the highest dose of procymidone was found. At the same mixture dose, the incidence of genital dysgenesis in the range-finding study [23] was around 40%, whereas the incidence in the present mixture study was around 17%. This

difference may be related to the lower number of litters in the groups in the range-finding study and hence greater uncertainty in results. The percentage of offspring with genital malformations was markedly higher in the highest mixture group than in any of the groups dosed with the individual pesticides alone at the dose included in the mixture (Table 8). The individual pesticides alone caused no statistically significant effect at this dose level. As such, these results show severe mixture effects at dose levels where the individual pesticides caused no effect when given alone. Similar results were found in another mixture study of three similarly acting anti-androgens, i.e. the AR antagonists flutamide, procymidone and vinclozolin [6]. Modeling was not done for genital malformations, because there were insufficient data for the single pesticides regarding this endpoint.

4.2. *In vitro* studies

The objective of performing the *in vitro* assays was to evaluate the predictability of *in vitro* methods for estimating *in vivo* developmental effects. The ranking of the potencies for AR antagonism was: mixture ~ procymidone > prochloraz ~ epoxiconazole > tebuconazole. As the content of procymidone in the mixture actually constituted only approximately 1/3 of the total mixture dose, these results indicate combination effects on this endpoint. This is in line with previous observations that chemicals act additively to antagonize AR [56].

Epoxiconazole, prochloraz and tebuconazole and the mixture were all shown to reduce testosterone and estradiol levels and increase progesterone levels in H295R cells. In general the effects of prochloraz was most pronounced on all three hormones. In regards to translating the *in vitro* results into *in vivo* effects in developmental rat studies, the lack of effects of mancozeb on steroid hormone synthesis *in vitro* fit well with the lack of observed anti-androgenic effects of mancozeb *in vivo*. Based on the *in vitro* results, one might also expect to see an effect on the measured hormone levels, particularly on the progesterone levels in the rats. In the present study, progesterone levels were unfortunately only investigated in the offspring at 50 days of age, and here no significant effects were seen [24]. However, results from previous *in vivo* studies with azole fungicides show that pregnant rats dosed with epoxiconazole (15 and 50 mg/kg bw/day) and tebuconazole (50 and 10 mg/kg bw/day) during gestation had a marked increase in progesterone levels in plasma at GD 21 [18].

Concerning the prediction of anti-androgenic effect (e.g. nipple retention), we would expect from the *in vitro* studies that procymidone and the three azole fungicides had the potential to induce this effect. Procymidone was the most potent AR antagonist *in vitro* and had also effects on nipple retention. Prochloraz was the most potent testosterone inhibiting chemical *in vitro* and was found to affect nipple retention in the present study and has previously been shown to reduce male AGD *in vivo*, although higher doses than those used in this project were necessary to induce this effect [53]. All taken together, the *in vitro* studies gave good indications of whether to expect *in vivo* anti-androgenic effects and effect on gestation length. Overall, application of predictable *in vitro* assays may contribute to faster and cheaper data generation for evaluation of chemicals and a reduction of the number of experimental animals that have to be used for risk assessment of chemicals.

4.3. Regulatory perspectives

In the present paper mixture effects of endocrine disrupting pesticides have been shown, and as such the results support that there is a need for risk assessment of the cumulative exposure to chemicals. Below the mixture results are discussed in relation to values used in regulatory risk assessment of single chemicals such

Table 9

LOAELs for effect on gestation length (GL) in dams and nipple retention (NR) in male offspring after exposure to the single pesticides alone or in mixture.

Pesticide	Epoxiconazole		Prochloraz		Tebuconazole		Procymidone	
	Single	Mixture	Single	Mixture	Single	Mixture	Single	Mixture
LOAEL-GL	15	2.5	>35	5.8	>50	8.3	–	–
LOAEL-NR	>15.0	1.3	5.8	2.9	8.3	4.2	4.2	4.2

Mancozeb and procymidone are not included for GL, and mancozeb is not included for nipple retention, as the pesticides did not affect these endpoints.

Table 10

ADI and derived no effect level (DNEL) in mg/kg bw/day for nipple retention in male offspring after exposure to the single pesticides alone or in mixture.

	ADI ^a	DNEL single	DNEL mix
Epoxiconazole	0.008	0.050	0.004
Prochloraz	0.010	0.019	0.010
Tebuconazole	0.030	0.028	0.014
Procymidone	0.003	0.014	0.014

DNEL is calculated as LOAEL divided by 300. Mancozeb is not included as this pesticide did not affect nipple retention.

^a The ADI values for the four pesticides respectively are from EC (2008a), FAO (2004), EC (2008b) and EFSA (2009).

as LOAELs, Derived No Effect Levels (DNELs) and Acceptable Daily Intake (ADI).

The observed LOAELs for the pesticides singly and the doses of the pesticides in the mixture at the mixture LOAEL were compared for the endpoints gestation length and nipple retention (Table 9). This was done only for the relevant pesticides in the mixture, *i.e.* those that singly caused effects on these endpoints. The main purpose was to estimate whether mixture exposure would lead to lower LOAELs than seen after single chemical exposure. For gestation length, the doses of the single pesticides at the mixture LOAEL were approximately 6 times lower than the LOAELs for the pesticides given alone. This is an important finding, as it further illustrates that mixture exposure leads to lower effect levels. For nipple retention, several of doses of the single pesticides at the mixture LOAEL were also lower than those for single chemical exposure. However, for procymidone, there was no difference between single and mixture LOAELs. Also, the differences between single and mixture LOAEL for prochloraz and tebuconazole was only 2 times, and not 6 times like it was seen for gestation length. It must, however, be kept in mind that the mixture caused effect on nipple retention at the lowest mixture dose studied, *i.e.* the mixture LOAELs may have become lower if lower doses of the mixture had been studied. LOAELs determined experimentally depend on many factors such as the selection of strain or species, the selection of doses levels, sample size, the end points assessed and biological variation. Thus some variability is to be expected among LOAELs and may range from less than twofold up to a factor of 10 [57]. However, the LOAELs in this study are from studies in the same strain of rats, the dose levels were carefully selecting based on the previous results in our laboratory and the same endpoints were assessed. Also, the results were quite reproducible for both the single pesticides and the mixture when comparing the results from our previous studies on the single pesticides, the range-finding study with the mixture and the large mixture study where both mixture doses and doses of the single pesticides were included ([23], Supplemental Material Table X1). Thus, the variation in our study LOAELs are clearly below a factor of 6 substantiating that this factor represents an increase beyond variation within and between studies.

In Table 10, the Acceptable Daily Intakes (ADIs) are compared to the DNELs for effect on nipple retention for each of the pesticides. The DNEL was calculated from the LOAELs shown in Table 9 using the normally used uncertainty factors for deriving ADIs, *i.e.* 10 for extrapolation from animals to humans, 10 for different

sensitivity among humans and 3 for extrapolation from an effect level to NOAEL. All of the DNELs for single chemical exposure to the pesticides were higher or similar to the ADIs, reflecting that the current ADI probably protects against the endocrine disrupting effect of the studied pesticides, when the pesticides are evaluated chemical by chemical. The mixture DNELs were, however, in most cases lower than the ADIs. This was especially the case for epoxiconazole and tebuconazole, where the DNELs based on the respective LOAELs were half of the ADI. This indicates that the ADIs are not sufficiently low to protect against the mixture effects of the pesticides studied in this project. An exception was procymidone where the ADI was lower than the mixture DNEL.

A survey of the dietary cumulative intake using the probabilistic approach did not show a reason for concern in relation to mixed exposure of Danish consumers to the five investigated pesticides (Jensen et al. [58]). The risk assessment included an additional uncertainty factor of 3 to allow room for human exposure to other endocrine disrupting chemicals. However, it is somewhat uncertain whether this factor is sufficient for covering the already known endocrine disrupters and especially the thousands of chemicals that are potential endocrine disrupters based on *in vitro* data and QSAR modeling [59,60].

4.4. Conclusions

Combined developmental exposure to endocrine disrupting pesticides at dose levels below NOAELs for the single pesticides caused adverse effects on male sexual development and gestation length and the mixture effects were well predicted by dose-addition at low doses.

The *in vitro* data showed that the AR reporter gene assay and the H295R steroidogenesis assay gave good indications of the observed *in vivo* effects. As humans may be exposed to thousands of chemicals for which we still have no data about endocrine disrupting properties, further *in vitro* studies as well as QSAR modeling are important for progress in this area. *In vitro* data are especially important for decisions on how to group endocrine disrupters for cumulative risk assessment.

Comparisons of the DNELs for the mixture effect on nipple retention to the Acceptable daily intake (ADI) indicate that ADIs are not sufficiently low to protect against the mixture effects of the pesticides shown in this study. Thus, the results imply that risk assessments based on NOAELs for single chemicals likely underestimate the risk and that there is a need for modification of risk assessment procedures for pesticides, in order to take account of mixture effects and the potentially serious impact of mixed exposure on development and reproduction. One possibility is to allow each pesticide to use only a fraction of the ADI, where the fraction should depend on the number of relevant endocrine disrupters in the specific exposure scenario. Another possibility is to base the cumulative risk assessment on dose-additivity modeling. Observations of similar type of effect are recommended as the basis for grouping chemicals for cumulative risk assessment as the results of our study support findings from previous studies showing dose-additive mixture effects of endocrine disrupters that induce similar types of effects, but have dissimilar mechanisms of action. Last but not least, it is recommended when considering cumulative

risk assessment to include all kinds of chemicals e.g. pesticides, industrial chemicals, environmental contaminants, as endocrine disruptors exist within all of these chemicals classes and humans may be exposed to several of them simultaneously.

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Appendix A. Supplementary data

Supplementary data associated with this article can be found, in the online version, at <http://dx.doi.org/10.1016/j.reprotox.2012.05.090>.

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